Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/05/2006
PI Name:	Li, Chuan-Yuan Ph.D.		
Project Title:	Mechanisms of HZE Particle-Induced Genetic Instability/Oncogenic Transformation and Their Prevention		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Radiation biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Comments:	PI moved to Duke University in December 2011. Formerly at University of Colorado Denver (Ed., 2/8/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-02
Start Date:	07/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2007 per A. Chu/ARC (6/07) NOTE: 10/2006--moved to U CO Health Sciences Center from Duke; new grant number given for UCHSC; research at Duke will continue		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1629		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this proposal is to carry out mechanistic studies of genetic instability/neoplastic transformation induced by HZE particles and to develop potential approaches to reduce/prevent it. We will initially focus our efforts on characerizing the phenomenon of HZE particle induced persistent genetic instability in mammalian cells. Next, we will evaluate the potential relationship between genetic instability and oncogenic transformation. Lastly we will explore the potential roles of oxidative stress in HZE particle induced genetic instability/oncogenic transformation and to evaluate if anti-oxidative agents can be effective countermeasures against genetic instability/oncogenic transformation. We have three objectives: 1) To characterize genetic instability induced by HZE particles in mammalian cells. This is to test the hypothesis that HZE particle-induced genetic instability is qualitatively similar to those induced by low-LET radiation. 2) To study the potential role of HZE particle-induced persistent genetic instability in oncogenic transformation of exposed cells. This is to evaluate the hypothesis that HZE particle-induced genetic instability will significantly enhance oncogenic transformation. 3) To study the role of oxidative stress in HZE particle induced genetic instability/oncogenic transformation. Both genetic and chemical approaches will be used to study the role of ROS(reactive oxygen species). This is to test the hypothesis that ROS plays major roles in HZE particle-induced genetic instability/oncogenic transformation, and the instability/transformation can be reduced/prevented by the suppression of ROS. The development of malignant diseases is one of the consequences of exposure to cosmic galactic rays of most concern. The experiments to be conducted in this project will evaluate the risks in this respect. Specifically, we will explore the relationhip among HZE particle induced oxidative stress, genetic instability, and oncogenic transformation. The results from this study will faciliate a better understanding of the oncogenic risks associated with exposure to cosmic galactic rays and provide directions for the development of countermeaures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed study will provide a comprehensive estimate of genetic instability induced by HZE particles and the potential involvement of reactive oxidative stress in this important process. It can potentially lead to countermeasures that can reduce/prevent the genotoxic and carcinogenic effects of HZE particle exposures suffered by astronauts. In addition, we will understand the relationship between genetic instability and oncogenic transformation in a clearer fashion, which will benefit the general population on earth.
Task Progress:	We have made the following progress so far. 1) Using standard micronucleus (MN) formation assay, we have obtained data indicating clear HZE particle induced increase in MN. On the other hand, the addition of anti-ROS agents such as a porphyrin-based superoxide dismutase mimetic MnTE-2-PyP5+ significantly($P<0.05$) reduced the rate of MN induction. Indicating that superoxide radicals play important roles in HZE particle induced DNA damage. 2) In addition to MN assays for evaluating DNA damage, we have established the following two assays that can monitor genetic instability more accurately: A. A GFP-based reporter that can detect either small intragenic mutations or deletions in the reporter gene. This assay should be very suitable for the rapid detection of acute or persistent genetic instability induced by HZE particles. B. The BigBlue reporter system that can accurately evaluate small intragenic deletions. The BigBlue system consists of bacterial phage-based reporter genes that stay silently in rodent (rat or mouse) genome. It is thus susceptible to genetic instability in the host genome. The major advantage is that the after exposure to mutagenic agents, there is no further culturing or expression period needed. Extraction of host DNA and packaging with commercially available packaging extract allow accurate and quantitative measurement of the mutation suffers in the reporter. As delayed type mutations have been associated with oxidative stress related mutations, we thought this system is well suited for our studies despite some concerns about its utility. Our preliminary results with the HZE particles proved this to be true. 3) We have made two runs of experiments (Fall and Spring) in 2005-2006 with (1GeV) Fe ions at NSRL facility located in BNL. Both experiments went through smoothly. 4) With these runs we have obtained the following preliminary results, with more repetitions necessary in the planning in the coming NSRL runs: A. In terms of delayed persistent genetic instability, we have data suggesting clear HZE-induced micronucleus formation in cells that were exposed to HZE particles more then 25 cell divisions earlier. B. In addition, when the GFP-based reporter system was used, significant persistent genetic instability was also observed in cells that were irradiated 25 cell division earlier. C. In addition to the GFP-reporter data, which mostly detect deletion-based mutations, we have also started to adopt the BigBlue system for our studies. The BigBlue reporter mostly picks up small intragenic mutations that are typical during persistent genetic instability. Our results using the rat MEF cell line with the BigBlue reporter gene indicated an elevated frequency of mutation mutations in the surviving clones. Of the 8 survivor colonies analyzed, the mutation frequency ranged from 2.0x10-5 to 2.0x10-4 compared with the control frequency of 8x10-6, indicating significant up-regulation of mutation rate by HZE particles in these cells. These are very exciting data since it gives us a quantitative handle on delayed genetic instability in the HZE-irradiated population. We plan to run more experiments, especially in the presence of anti-ROS agents such as SOD mimetic and NAC. D. We have developed a novel GFP-based mismatch repair system where mutations in microsatellite sequences can be detected by EGFP fluorescence through either flow cytometry or fluorescence microscopy. We have used this system to evaluate X-ray induced mutagenesis and preliminary results indicate the effectiveness of the approach. We plan to use this system to evaluate if microsatellite mutations are significantly affected by HZE particles.
Bibliography Type:	Description: (Last Updated: 10/30/2019)
Articles in Peer-reviewed Journals	Yan B, Wang H, Peng Y, Hu Y, Wang H, Zhang X, Chen Q, Bedford JS, Dewhirst MW, Li CY. "A unique role of the DNA fragmentation factor in maintaining genomic stability." Proc Natl Acad Sci U S A. 2006 Jan 31;103(5):1504-9. PMID: 16432220 , Jan-2006
Articles in Peer-reviewed Journals	Yan B, Wang H, Wang H, Zhuo D, Li F, Kon T, Dewhirst M, Li CY. "Apoptotic DNA fragmentation factor maintains chromosome stability in a P53-independent manner." Oncogene. 2006 Apr 17; [Epub ahead of print] PMID: 16619042 , Apr-2006